

$^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$ **2020Ke03,2021Ke04**

2020Ke03,2021Ke04: a 95-MeV ^{24}Mg beam was produced from the ATLAS accelerator at ANL and bombarded a $\approx 200\text{-}\mu\text{g}/\text{cm}^2$ ^{12}C target. Prompt γ rays were detected using the GREYINA tracking array consisting of 12 modules. Fusion-evaporation residues were separated and identified by the Fragment Mass Analyzer (FMA) with a focal-plane position-sensitive parallel-grid avalanche counter (PGAC) and an ionization chamber. Measured E_γ , I_γ , $\gamma\gamma$ -coin, recoil- $\gamma(\theta)$. Deduced levels, J , π . Compared with mirror nucleus ^{34}S and shell model calculations. Calculated astrophysical $^{33}\text{Cl}(p,\gamma)^{34}\text{Ar}$ reaction rate and discussed its implications for the identification of nova presolar grains from sulfur isotopic abundances.

 ^{34}Ar Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$E(\text{level})^\dagger$	$J^\pi^\ddagger$
0.0	0^+	4020.6 13	2^+	4851.9 8	3^+
2091.5 4	2^+	4131.6 9	1^+	4881.4 12	2^+
3289.1 5	2^+	4517.5 7	3^-	4963.7 13	0^+
3876.4 9	0^+	4644.1 9	4^+	4966.9 11	2^-
				5061.8 8	1^+

† From a least-squares fit to γ -ray energies.

‡ As given in **2020Ke03** and **2021Ke04** based on known assignments of low-lying levels, comparison with mirror nucleus ^{34}S , characteristic γ -decay patterns, and $\gamma(\theta)$ analysis. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

 $\gamma(^{34}\text{Ar})$

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult.	Comments
832.1 9		1.6 2	4963.7			0^+	4131.6 1^+
842.5 7		2.4 2	4131.6			1^+	3289.1 2^+
1197.5 4		37.9 16	3289.1			2^+	2091.5 2^+
1228.4 5		15.8 7	4517.5			3^-	3289.1 2^+
							D Mult.: $\Delta J=1$, dipole. $A_2=-0.35$ 10, $A_4=+0.03$ 12.
1562.8 7		5.8 4	4851.9			3^+	3289.1 2^+
1592.5 17		1.4 2	4881.4			2^+	3289.1 2^+
1771.8 11		5.0 4	5061.8			1^+	3289.1 2^+
1784.8 8		5.9 4	3876.4			0^+	2091.5 2^+
1930.4 23		4.8 9	4020.6			2^+	2091.5 2^+
2091.3 5		100 5	2091.5			2^+	0.0 0^+
2424.7 22		2.3 5	4517.5			3^-	2091.5 2^+
2552.4 8		25.1 16	4644.1			4^+	2091.5 2^+
							Q Mult.: $\Delta J=2$, quadrupole. $A_2=+0.20$ 8, $A_4=-0.01$ 11.
2759.9 12		11.8 9	4851.9			3^+	2091.5 2^+
2788.9 19		4.4 5	4881.4			2^+	2091.5 2^+
2875.2 10		11.6 9	4966.9			2^-	2091.5 2^+
							D Mult.: $\Delta J=0$, dipole. $A_2=+0.9$ 3, $A_4=+0.4$ 5.
3289.1 10		5.3 6	3289.1			2^+	0.0 0^+
4019.8 15		7.4 9	4020.6			2^+	0.0 0^+
4881.9 24		3.6 6	4881.4			2^+	0.0 0^+
5062 ‡		<1.4	5061.8			1^+	0.0 0^+

† From **2020Ke03**.

‡ Placement of transition in the level scheme is uncertain.

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Legend

Level Scheme

Intensities: Relative I_γ

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$
 $-\cdots-\cdots$ γ Decay (Uncertain)

